

Midterm Bonus Assignment

Ewan Pedersen

November 11, 2024

Abstract

This document serves as the solution to the Midterm Bonus Assignment problems.

1 Introduction

Euler showed that the sum of the inverses of the squares of the positive integers converges to $\pi^2/6$:

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The purpose of this assignment is to prove this fact using methods from single and multivariable calculus.

2 Problem 1

Prove that the double integral and the infinite series shown below are equal:

$$\int_0^1 \int_0^1 \frac{1}{1-xy} dx dy = \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

3 Main Results

4 Conclusion